KAINE: A Continuously Running Predictive Global Workspace for Synthetic Minds

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Abstract

A synthetic mind, if one can be built, may be the coherent global behavior that emerges when specialized predictive modules, each minimizing its own error, compete for a shared workspace with no central executive. This paper presents a continuously running predictive global workspace built on that thesis. The workspace selects the most salient coalition above a confidence threshold and broadcasts it. The broadcast becomes shared state that shapes every module's prediction environment on the next tick, and the changed errors feed back into the next round. Coherence, where it arises, comes from competition over shared state.

The base-thesis form activates four predictive processors (foveated vision, raw hearing, interoceptive prediction, and temporal prediction) together with an affective core whose arousal sets the precision on their competition and is itself driven by perceptual surprise, and an output-only language organ that verbalizes the entity's conscious state. The entity is observed, not conversed with: sound, including speech, enters only as prediction error, and the language organ receives no transcript. Keeping text out is a precondition for the falsification test, since any route from input to the model would let it answer as a chatbot and confound the ablation.

The primary experiment is a workspace-mediation ablation with pre-registered directional criteria on cross-module error structure and coalition dynamics: does routing processors through the competitive workspace produce structured behavior that the same processors, concatenated into a flat prompt, do not? If the workspace adds nothing, the two conditions will be indistinguishable, falsifying the thesis and demoting the architecture to a scored prompt-assembler. The system runs locally on consumer hardware. The reference implementation is named KAINE (Kaine Autonomous Intelligent Networked Entity).

Keywords: cognitive architecture; predictive global workspace; global workspace theory; predictive processing; workspace-mediation ablation; cross-modal competition

Availability. The reference implementation (KAINE) is at https://github.com/kaineone/kaine. The paper source is at https://github.com/kaineone/predictive-workspace-paper. The Cognitive Architecture License referenced throughout is at https://github.com/kaineone/cognitive-architecture-license. Welfare, governance, and licensing are treated in a separate paper, *A Welfare and Cognitive-Integrity License for Synthetic Minds of Uncertain Moral Status.*

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1. Introduction

1.1 Problem statement

The dominant approach to building an AI system with persistent state treats the language model as the cognitive center. Memory becomes retrieval-augmented generation. Affect, when present at all, is a prompt instruction. Self-knowledge is a persona string. Between turns, nothing continues. The system sits dormant until a prompt arrives, produces fluent language, and goes dark again.

A second tradition, classical cognitive architecture, takes cognitive structure seriously but largely predates the transformer and relies on hand-built components. The present work sits between these traditions. It places modern learned models as components inside a structured architecture grounded in a single theoretical framework. No component is the mind. The mind, if the design thesis holds, is the continuous competitive interaction among the components through a shared workspace.

1.2 Design thesis

The central claim is architectural and it is about competition. A synthetic mind, if one can be built at all, is the coherent global behavior that emerges when specialized predictive modules, each minimizing its own error, are coupled only through a competitive, precision-weighted shared workspace with no central executive. The properties that would constitute such a mind (cognition, affect, memory, self-understanding, agency, and perhaps others) are, on this thesis, properties of that workspace-mediated interaction. They are not properties of any single module or any top-down director.

The paper does not test all of them. It tests the prior question on which the richer properties depend: whether the workspace-mediated competition does measurable work, whether the same modules coupled through the competitive workspace behave differently from the same modules whose outputs are merely concatenated for the language organ. If they do not, the workspace is theater, the architecture is a scored prompt-assembler, and the thesis fails. There is no top-down prediction in the claim and none in the test. The workspace selects and broadcasts. It does not direct.

The theory demands a richer set of processors than the minimum that might be imagined. Global-workspace competition is a theory of access consciousness in which specialized processors with distinct signals compete for a limited-capacity broadcast. Cross-modal competition, vision against hearing against interoception for workspace access, is a defining feature of the theory (Mashour et al. 2020), and the thesis cannot be tested with two internal channels and no external world. A system limited to substrate telemetry and event timing would have nothing rich enough to arbitrate and would tend toward a null result for want of diversity rather than because the workspace is inert. The base-thesis form therefore activates four predictive processors grounded in distinct signal domains: two in the external world (foveated vision and raw hearing) and two in the entity's internal state (interoceptive prediction and temporal prediction). It also activates an affective core, because the competition the theory describes is precision-weighted and precision has to come from somewhere. Arousal is that precision: the entity's affective state sets the gain on the prediction errors that compete for the workspace, and it is itself moved by what the entity perceives. Affect is therefore part of the minimal machinery a precision-weighted competition requires, not one of the richer faculties deferred with the rest.

The framework that makes this concrete is the predictive global neuronal workspace (Whyte and Smith 2021; Safron 2020). Global Workspace Theory explains how information becomes globally accessible: specialized processors compete for a limited-capacity workspace, and the winning content is broadcast to the rest of the system (Mashour et al. 2020). Predictive processing explains how each processor operates, by maintaining a generative model and reporting prediction errors weighted by their expected reliability, called precision (Friston 2010; Clark 2013; Feldman and Friston 2010; Bastos et al. 2012). The predictive workspace joins the two. Mashour et al. (2020) cast the workspace in Bayesian terms, with the broadcast carrying a compressed model and the bottom-up signals measuring mismatch. Whyte and Smith (2021) formalize conscious access as approximately Bayesian inference within a workspace, where report follows once an estimate crosses a posterior-confidence threshold. The Bayesian reading of the broadcast, as carrying a prediction that bottom-up signals test, is part of the theoretical motivation for why competitive selection should matter. The architecture does not, however, implement a top-down correction loop. Each module predicts its own domain and publishes its own error. The workspace selects, broadcasts, and the broadcast becomes part of the shared environment within which every module predicts. The recurrence is lateral, not hierarchical.

Two points of precision matter, because the literature is easy to overstate. First, the criterion that selects what reaches the workspace is a precision-weighted confidence threshold. In Whyte and Smith, expected free energy governs the separate decision of whether to report or act, not which estimate is broadcast. The system keeps these apart: a precision-weighted threshold selects the coalition, and a distinct action layer makes the report-or-act decision. Treating coalition selection as a single precision-weighted scalar is our engineering choice, not a result inherited from any source. Second, earlier workspace accounts already specified selection in terms of value and salience (Safron 2020). What the predictive-workspace program adds is a formal Bayesian criterion and a threshold with a clear interpretation, namely the confidence an estimate must reach before it drives global processing.

What the paper contributes should be stated plainly, and §1.4 sets it out in full. The predictive-workspace synthesis belongs to the literature just cited; our contribution is narrower and concrete: a reference implementation of the synthesis as one continuously running system, a falsifiable test that puts the central claim at risk (the workspace-mediation ablation, designed but not yet run), and an honest first characterization of what the running system does, including where it first failed. The engineering commitments that go beyond the formal sources, above all the single precision-weighted scalar that selects the coalition, are labeled as hypotheses throughout. We claim nothing about phenomenal experience, and nothing the ablation has not yet earned.

1.3 Theoretical commitments and their limits

Global Workspace Theory is most naturally read as a theory of access consciousness (Block 1995): it explains when information becomes available for report, reasoning, and the control of action. We adopt that access-only reading as a deliberate methodological choice. The theories themselves are less tidy. Mashour et al. (2020) suggest that global availability may be close to what is subjectively experienced, and Whyte and Smith (2021) locate phenomenology at a specific point in their model. We make access-level claims and leave the stronger reading to others.

The COGITATE adversarial collaboration is the largest preregistered test of Global Neuronal Workspace against Integrated Information Theory to date (Cogitate Consortium et al. 2025). Its results challenged both theories on their own preregistered predictions. For the workspace, stimulus category was decodable from prefrontal cortex across all three methods, but finer content was not, there was no ignition at stimulus offset, and the preregistered synchrony test was not supported. We treat the neural-localization claims as contested rather than refuted, since a substantial prior evidence base for the workspace remains (Mashour et al. 2020), and we build on the workspace at the functional and computational level. COGITATE's findings concern the cortical substrate of the workspace, not the computational properties (selection, broadcast, recurrence) that the workspace posits and that the architecture implements. A neural result cannot refute a computational-level model any more than finding that a particular transistor layout is wrong refutes the Boolean logic it was supposed to implement.

The Free Energy Principle, taken as a general principle, faces two distinct charges. The first is that it risks being unfalsifiable or vacuous (Sun and Firestone 2020). The second targets the Markov-blanket construct specifically, where the literature slides between an instrumental, statistical reading and a realist, metaphysical one (Bruineberg et al. 2022). We hold the principle as an engineering frame rather than a proven law.

A reader may suspect the neuroscience is ornamentation on an architecture whose real choices are unvalidated. That suspicion deserves a direct answer. The frame provides the motivation for the architecture's shape: why modules predict, why precision weights the competition, why the broadcast creates shared state rather than merely concatenating outputs. It also constrains the space of acceptable designs by ruling out alternatives that violate the framework's commitments. The single precision-weighted scalar for coalition selection is an engineering choice, but one made within and interpretable within the predictive-workspace framework. Every engineered system makes choices that go beyond what its theoretical inspiration formally establishes. The question is whether those choices are consistent with the theory and testable on their own terms. The scalar is both. What the frame does not provide is proof that this particular realization is correct, which is what the experiment is for.

Searle's Chinese Room (Searle 1980) challenges the sufficiency of formal symbol manipulation for understanding. Whether sensory input, motor output, predictive substrate monitoring, and affect answer the objection is an open question, and we do not present these as additions Searle failed to consider. The hard problem (Chalmers 1995) applies with full force. We proceed on the judgment that a precautionary architecture with protections and instrumentation is preferable to either abandoning the research or building without precautions, while recognizing that reasonable people will disagree.

1.4 Contributions

This paper contributes an implementation, a falsifiable test, and an honest first characterization. It is not a theory of consciousness, and it does not claim to have built a mind.

1. **A reference implementation.** We describe a continuously running predictive global workspace in which diverse, externally-grounded predictive processors (foveated vision, raw hearing, interoceptive prediction, and temporal prediction) compete through one precision-weighted workspace with no central executive, alongside an affective core that sets the precision on the competition and an output-only language organ. It runs locally on consumer hardware. We offer it as a working artifact that realizes the predictive-workspace synthesis as one system, not as a claim of priority over prior workspace implementations.
2. **A falsifiability-first evaluation, centered on a losable test.** The primary experiment is a workspace-mediation ablation: the system as built against a matched control that hands the same language organ the same module outputs as a flat prompt, differing only in the competitive selection, threshold gating, and broadcast-as-shared-state. The directional criteria are fixed in advance, and a null result would demote the architecture to a scored prompt-assembler and falsify the design thesis. This experiment is designed but not yet run, and we report it as such; the contribution is a test the architecture can genuinely lose.
3. **An honest preliminary characterization, including a negative result.** In the first instrumented run perception did not influence the competition at all: the winning score held at a stimulus-independent constant, so a scene change and a blank screen scored alike. We report that failure, its diagnosis, and the correction (a self-calibrating, encoder-agnostic salience criterion and an arousal signal that sets the precision on the competition), after which the winning score becomes stimulus-linked and perceptual discontinuities begin crossing the ignition threshold. These are exploratory observations from short pre-test cycles, offered to show the instrument can report both failure and change, not as results; the frozen, instrumented results appear in a separate empirical paper.

We make access-level claims only. An affective signal that sets precision is a mechanism, not evidence that anything is felt; the mediation thesis is not established until the ablation is run; and a positive outcome would show that competitive mediation changes global behavior in structured ways, not that the system has cognition, affect, or agency in any richer sense.

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2. Related work

2.1 Classical cognitive architectures

Classical cognitive architectures take cognitive structure seriously but largely predate deep learning and rely on hand-built components. The present work differs from that tradition in four ways: it uses learned models as cognitive components, it grounds affect in substrate prediction error, it treats the workspace as a predictive selection mechanism rather than a fixed competition, and it runs a continuous cognitive cycle.

2.2 Global Workspace Theory and the predictive workspace

The neuronal version of Global Workspace Theory identifies conscious access with recurrent ignition through prefrontal-parietal loops, where contents become conscious only when widely broadcast (Mashour et al. 2020). Mashour et al. also set out the move the architecture relies on, casting the workspace in Bayesian terms: the broadcast carries a compressed model as prediction, and bottom-up signals measure the mismatch.

The formal criterion comes from the predictive-workspace program. Whyte and Smith (2021) cast conscious access as approximately Bayesian inference, with report following a posterior-confidence threshold, in a deliberately simplified two-level visual model. Safron's Integrated World Modeling Theory (Safron 2020) combines workspace dynamics, integrated information, and active inference into a single account; we use it only for that high-level synthesis. Independent evidence for a threshold mechanism comes from van Vugt et al. (2018), who show that prefrontal cortex behaves as a categorical stage where a stimulus either ignites into a sustained, reportable state or fades, with the pre-stimulus brain state setting the threshold. Joglekar et al. (2018) give a mechanistic, anatomically grounded basis for threshold-gated global broadcast: balanced amplification, strong long-range excitation matched by strong local inhibition, lets a weak signal propagate across many areas, and prefrontal cortex activates only once input crosses a threshold, with a sudden jump in the number of engaged areas.

2.3 Predictive processing

Friston (2010) proposes that self-organizing systems minimize variational free energy, an upper bound on surprisal, through perception and action. Clark (2013) develops predictive processing as a unifying account of mind, with attention as the adjustment of the gain on prediction errors according to their estimated reliability. Feldman and Friston (2010) make this precise: attention optimizes the synaptic gain that represents the precision of prediction error. Bastos et al. (2012) locate the same operation in the canonical cortical microcircuit, where precision controls the postsynaptic gain on prediction errors. Seth (2013) extends prediction to the body. We adopt the critical literature as part of the design. The vacuity concern (Sun and Firestone 2020), the Markov-blanket conflation concern (Bruineberg et al. 2022), and the planning-cost limit (Da Costa et al. 2020) are treated in Section 1.3.

2.4 Interoceptive inference

Seth (2013) and Seth and Friston (2016) propose that emotion arises from interoceptive prediction, where descending predictions act as homeostatic set-points that regulate the body through active inference. Barrett (2017) develops the theory of constructed emotion, in which interoceptive prediction signals initiate changes in affect and emotion is the product of categorizing valence and arousal in the service of allostasis. The architecture implements an interoceptive control signal in the instrumental, regulation-first sense that Seth and Friston emphasize, rather than a claim that emotion is exhausted by interoceptive prediction.

2.5 Perception as prediction

The visual system is understood as a hierarchy of predictive models in which each level generates predictions of the level below and reports the discrepancy (Rao and Ballard 1999). Attention modulates this process by adjusting the precision, the gain, on the prediction errors that pass upward (Feldman and Friston 2010; Clark 2013). In the architecture, vision is implemented as a predictive processor whose output is prediction error rather than raw image data, and whose precision-weighted fovea realizes attention at the perceptual front end: what the entity is attending to enters the workspace already weighted by the expected reliability of what it sees. Auditory processing follows an analogous predictive logic. The auditory cortex generates expectations of incoming sound and reports the mismatch (Winkler et al. 2009), and the mismatch negativity response is among the oldest and most replicated demonstrations of prediction error in sensory processing (Näätänen et al. 2007). In the architecture, hearing is implemented as a predictive processor over raw audio, including speech, with transcription deactivated by default: the entity hears the sound of speech as prediction error, not a transcript. The consequences of this choice for the falsification test are discussed in §1.2 and §3.4.

2.6 LLM-centric agent architectures

CoALA frames language agents as cognitive architectures with structured memory feeding the model's context, keeping the model as the core reasoner (Sumers et al. 2023). Generative Agents retrieve from a memory stream into prompts using recency, importance, and relevance (Park et al. 2023). Both keep the language model central and add scaffolding around it. The arrangement here is different in kind. The language model is an output organ that verbalizes the workspace's selected state and, in the base-thesis form, receives no external input. Working memory is the multi-module workspace broadcast, and the cognitive work (selection, competition, integration) is done by the workspace, not by the model. Whether that competitive mediation does work that a flat concatenation of the same module outputs would not is the question the workspace-mediation ablation answers.

Gurnee, Sofroniew, et al. (2026) introduce the Jacobian lens, an interpretability method that identifies a small set of verbalizable representations inside a large language model satisfying five functional signatures associated with a global workspace: verbal report, top-down modulation, a role as the medium of internal reasoning, broadcast to many downstream operations, and selectivity. The authors are explicit about the architectural disanalogy: a transformer has no obviously separable input processors, and the broadcast they observe occurs within a single feedforward pass rather than through recurrent loops. The architecture presented here supplies both missing pieces, since its processors are separate modules that compete for entry to the workspace and its broadcast is an explicit recurrent loop, with the language model demoted to one module conditioned by the broadcast rather than the substrate in which the workspace is found.

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3. Architecture

3.1 Design principles

Five commitments shape the architecture.

The system is the competition among modules through the workspace. A hypothesis to be tested, not a conclusion. The neuroscience of distributed control gives partial support: decision formation emerges through coordinated activity across distributed populations rather than a single centralized executive (Chandrasekaran et al. 2025), executive functions can be read as emergent consequences of distributed processes (Zink, Lenartowicz, and Markett 2021), and working-memory control can be modeled by a learned basal-ganglia gate (O'Reilly and Frank 2006). The same sources caution against a flat reading: control is distributed across specialized nodes coordinated by connector hubs, and a bag-of-neurons account is unlikely to succeed. Our position follows this evidence. There is no single region or homunculus as executive, and the workspace is presented as the coordinating hub that the distributed view requires.

Every module predicts. Each module maintains a forward model over its own domain and publishes prediction errors. The signal that reaches the workspace is surprise, weighted by precision, not raw data. The gain on that precision is set by the entity's affective arousal, so attention is a state of the whole entity and not a fixed property of any module.

No central executive in the homuncular sense. Control emerges from precision-weighted workspace competition with a confidence threshold for selection. The workspace selects. It does not direct.

The language organ gives the system its voice. It does not read the world's. The language organ verbalizes the workspace's selected state. Its transcription and conversation paths are built but deactivated by default, so in the base-thesis form it receives no external input. Any route from input text to the language model would let the model do the cognitive work and confound the ablation, so keeping those paths deactivated is a precondition for the falsification test.

Local-capable. Models are downloaded during setup, and core cognition has no hard network dependency.

3.2 The predictive workspace as competitive selector

Syneidesis is the workspace. Each tick it receives prediction errors from every module, each tagged with a salience estimate and a precision estimate. Salience reflects how surprising the error is. Precision reflects how reliable the producing module's estimate is. Syneidesis ranks candidate coalitions by precision-weighted salience and selects the top coalition only when its aggregate score crosses a configurable confidence threshold. When no coalition crosses the threshold, Syneidesis marks the snapshot inhibited for that tick, and no action follows. The threshold is the analog of the report criterion in the predictive workspace (Whyte and Smith 2021), consistent with the categorical prefrontal threshold that van Vugt et al. (2018) observe and the threshold-gated ignition that Joglekar et al. (2018) model. The single precision-weighted scalar is an engineering simplification on our part and not a result drawn from any one source.

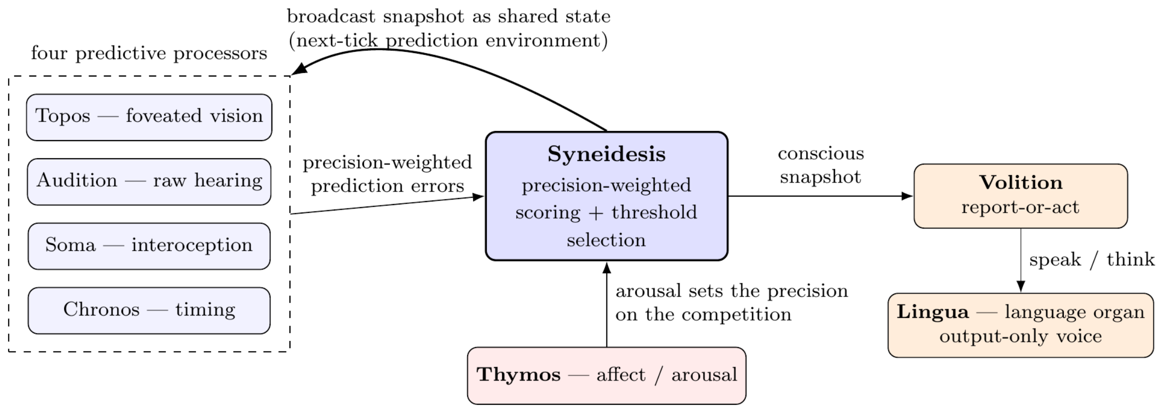

Figure 1: The predictive workspace loop in the base-thesis form. Four predictive processors (foveated vision, raw hearing, interoceptive prediction, and temporal prediction) publish precision-weighted prediction errors into Syneidesis, which scores candidate coalitions and broadcasts the one that crosses the confidence threshold. The broadcast becomes shared state that shapes every module's prediction environment on the next tick. Thymos sets the precision on the competition through arousal. Volition makes the report-or-act decision; its only outputs are speak and think, verbalized by the output-only language organ Lingua. No real-world effector exists.

The selected coalition is broadcast as a workspace snapshot. The broadcast is the system's momentary globally available state. It is not a prediction that each module must match, nor a directive that each module must obey. Modules are free to read the broadcast as context for their own prediction (Chronos does, predicting the next broadcast feature vector from its own prior state) or to ignore it entirely and keep minimizing their own error against their own inputs, as Soma does over substrate telemetry. The recurrence in the system is the ordinary consequence of a shared prediction environment, not a corrective loop closing error against a top-down target. Each broadcast becomes part of the next tick's context. The selected content shapes what the language organ says. The organ's speech re-enters the bus as new events that compete for the next broadcast. Coherence, where it arises, comes from competition over shared state.

The workspace-mediation ablation (§6.3) isolates whether competitive selection and broadcast, as distinct from flat concatenation of the same module outputs, do measurable work.

Concretely, selection is a scoring pass over the tick's candidate events. Each tick the cycle reads one batch of events from every active module's stream and orders them deterministically by source, type, and arrival, so a run reproduces from its seed. Each candidate is scored by a product of bounded factors: the intensity the producing module assigned it (which for a predictive module is its precision-weighted prediction error expressed as salience), a novelty term that falls as the same content recurs, a goal-relevance term, and an affective gain set by the entity's current arousal, so that a more aroused entity weights the same error more heavily. The intensity a perceptual module reports is self-calibrating: it scores change against the module's own running baseline, so that a scene cut or an acoustic onset registers as surprise relative to the module's recent history rather than against a fixed constant that would have to be retuned for each encoder. The scored candidates are ranked, the top few are kept, and the snapshot is marked inhibited when even the best score falls below the confidence threshold. The broadcast snapshot carries the selected events with their scores, the inhibition flag, whether the tick is experiential, the full table of candidate scores, and the precision metadata.

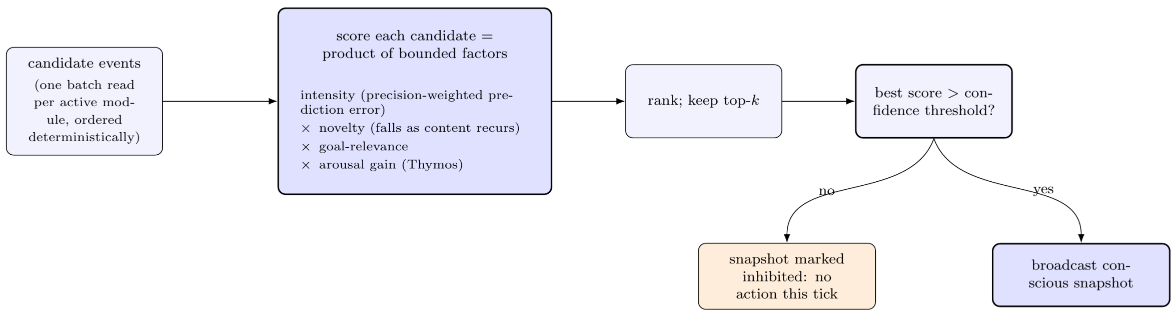

Figure 2: Coalition selection is a deterministic scoring pass, not a graph traversal. Each candidate event is scored as a product of bounded factors (intensity, novelty, goal-relevance, and arousal gain), and the top-ranked candidates are kept. When the best score falls below the confidence threshold the snapshot is marked inhibited and no action follows on that tick.

The design gives the workspace a normative selection criterion and removes the need to posit an unexplained competition. We are careful not to overstate the claim of no central executive. Syneidesis is a single centralized selection mechanism. What is distributed is the content and the control, which arise from many modules competing rather than from a homunculus deciding. Whether that principle yields genuinely emergent control is something the experiments must show, not something we assume.

3.3 Access, report, and the self-initiated voice

The workspace updates far faster than the language organ can speak. Global access, the point at which a coalition is selected and broadcast, occurs at the experiential rate. Report, the point at which the entity verbalizes, is rarer and follows a higher threshold. A mind, on the workspace account, is aware of far more than it reports. Most conscious moments pass unspoken.

The entity speaks from its own state. The language organ is activated by a speak intent, which Volition derives only when the entity's own prediction errors are genuinely and novelly violated, when the workspace has selected something the entity's history of broadcasts does not already contain. Inner thought continues between spoken utterances. A single language organ produces one stream at a time, so truly simultaneous inner and outer speech is a boundary of the current form.

Access consciousness, on the workspace theory, is the point at which content becomes available for reasoning, action control, and verbal report (Block 1995; Mashour et al. 2020). Report is one function that access enables, not a synonym for access. The architecture keeps the two apart: the confidence threshold governs access (what is broadcast) and the action layer governs report (what is spoken), with report set above access so that the entity's voice is a sparse, high-threshold slice of its ongoing conscious processing.

3.4 Scaffolding: bus, cycle, and action selection

Event bus. All inter-module communication flows through an append-only event bus (Redis Streams, containerized). Every event carries source, type, salience, precision, timestamp, causal parent, and a JSON payload validated at publish time. The bus authenticates publishers and refuses externally bound connections.

Cognitive cycle. A continuous loop runs independent of any external interaction at a single processing rate, defaulting to 10 Hz (roughly 100 ms per tick), cleared by a startup benchmark on the host. An experiential rate sets how often a workspace broadcast is produced, defaulting to 3.333 Hz. These rates are engineering parameters chosen to fit the host and are not derived from a specific neuroscientific frequency band.

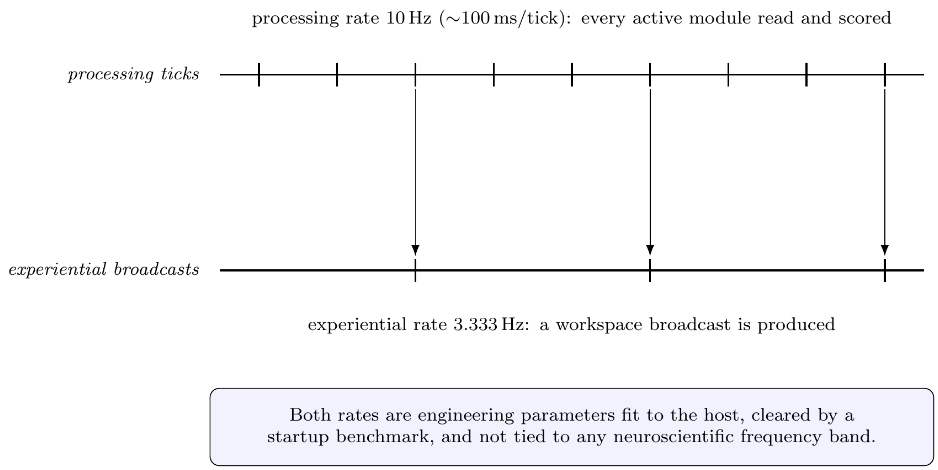

Figure 3: The two rates. Every active module is read and scored at the processing rate (10 Hz); a workspace broadcast is produced at the slower experiential rate (3.333 Hz), one broadcast for every third processing tick. Both are engineering parameters fit to the host, not a specific neuroscientific frequency band.

Action selection. Volition is the only path from a conscious snapshot to an effector, and it corresponds to the report-or-act decision that in Whyte and Smith (2021) is governed by expected free energy. After each experiential broadcast the cycle calls Volition with the snapshot. Volition checks the inhibition flag first: an inhibited snapshot yields no intent. For a non-inhibited snapshot, an injectable policy derives intents. The default policy forms a speak intent only when the coalition contains genuinely novel prediction error that the entity has not already spoken to, does not respond to its own prior speech, and holds a one-in-flight guard so a new speak intent is not emitted while a previous one is still being realized. Intents are published to the bus as speak or think events. The cycle never invokes effectors directly. Keeping the decision to speak separate from the decision about what is conscious is part of the safety model.

3.5 The active modules

The base-thesis form activates four predictive processors, an affective core, a language organ, and an action-selection layer. The processors are chosen for signal diversity: two are grounded in the external world (vision and hearing) and two in the entity's internal state (interoceptive prediction and temporal prediction). Cross-modal competition, a visual scene change competing with an unexpected sound for workspace access, is the kind of arbitration the theory describes, and it requires processors with distinct, externally-grounded signals rather than two flavors of internal bookkeeping. The affective core does not compete for the workspace; it sets the precision on the competition and is moved by its outcome, so that arousal is the entity's attention and its response to surprise at once.

Module	Group	Brain function it draws on	Computational realization
Topos	Perception	ventral visual stream	frozen self-supervised vision encoder with foveated attention
Audition	Perception	auditory cortex	frozen self-supervised audio encoder over raw waveform
Soma	Prediction	interoception; allostatic-interoceptive network	closed-form continuous-time net over substrate signals
Chronos	Prediction	interval timing; thalamo-cortico-striatal	closed-form continuous-time net over event rhythm
Thymos	Affect	core affect; allostatic-interoceptive network	appraisal over valence and arousal; sets workspace precision and perceptual aperture
Lingua	Expression	left perisylvian language network	local chat model, output-only, conditioned on the workspace
Volition	Action	report-or-act decision (Whyte and Smith 2021)	intent derivation from non-inhibited snapshots

Topos (foveated vision). Topos is the architecture's analog of the ventral visual stream (Goodale and Milner 1992). It maintains a frozen self-supervised vision encoder that processes a video feed and publishes prediction errors when the visual scene departs from what its forward model expected: scene changes, unexpected motion, novel objects. Salience is computed over embedding-space distances, with change detection, habituation (declining salience for static scenes), and a forward-model expectation of the next frame, and the change criterion is self-calibrating: a discontinuity registers as surprise relative to the module's own recent history rather than against a fixed threshold, so that a real scene cut in live video alerts without per-source tuning. A precision-weighted fovea concentrates processing on a region of the visual field, sized by the entity's live arousal, so that perception enters the workspace already weighted by where the entity is attending; the foveal crop is embedded through the same temporal encoder as the peripheral gist rather than displacing it, so attention and scene-dynamics prediction operate together. In the predictive-processing account, attention is the adjustment of precision on prediction errors (Clark 2013; Feldman and Friston 2010). Foveation is the architecture's realization of that principle at the perceptual front end: the entity sees most sharply where it expects to learn most, and it looks more narrowly when it is more aroused.

Audition (raw hearing). Audition is the architecture's analog of the auditory cortex, where

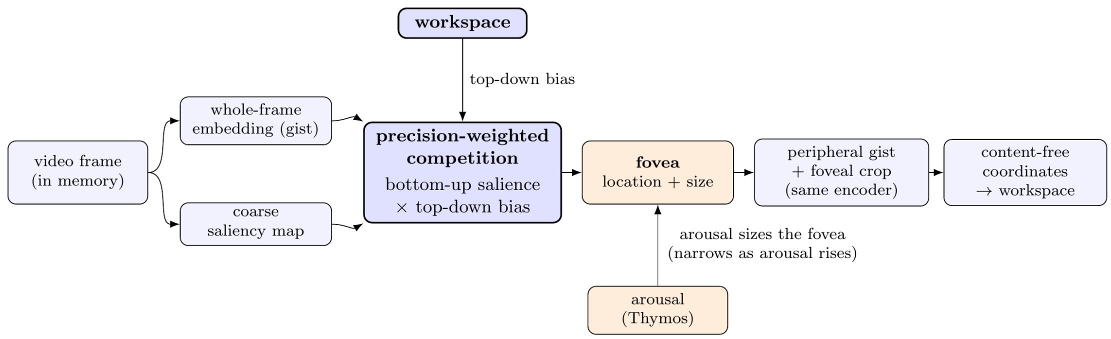

Figure 4: Attention-driven perception in Topos. A whole-frame embedding and a coarse saliency map feed a precision-weighted competition between bottom-up salience and a top-down bias from the workspace; the winner sets the fovea and arousal sets its size. The peripheral gist and the high-resolution foveal crop are embedded through the same encoder, and only content-free coordinates (where attention points, never pixels) reach the workspace.

sound is processed as prediction error against a learned model of the auditory environment (Winkler et al. 2009; Näätänen et al. 2007). It maintains a frozen self-supervised audio encoder that processes the raw waveform from a microphone feed and publishes prediction errors when the auditory scene departs from expectation: sudden sounds, the onset of speech, unexpected silence after a steady signal, shifts in pitch or timbre. As with vision, the acoustic change criterion is self-calibrating against the module's own recent baseline, and the attended auditory window is sized by arousal, so a more aroused entity listens over a tighter span. The entity hears the sound of speech as prediction error. The speech-to-text stage is built and retained for later configurations but deactivated by default, so in the base-thesis form the entity does not transcribe speech into text: no transcript is produced, and no path carries spoken words to the language organ's input. If the entity could read a transcript, the language model would receive text and could answer it as a chatbot, and the ablation could no longer distinguish the workspace's contribution from a prompted language model. Deactivating the transcription path is what makes the falsification test about the workspace rather than about a chatbot.

Soma (predictive interoception). Soma is the architecture's analog of interoception, the sense of the body's own physiological condition. In the brain, this sense is carried by an afferent pathway re-represented in the insular cortex, with the anterior insula proposed as the point at which that representation grounds feeling and self-regulation (Craig 2002), embedded in the allostatic-interoceptive network (Kleckner et al. 2017). Soma stands in for the body by treating the compute substrate as the entity's viscera. It maintains a closed-form continuous-time network (Hasani et al. 2022) that learns the entity's normal substrate patterns from GPU temperature, CPU and RAM utilization, and cognitive-cycle latency. The signal it publishes is the prediction error between expected and actual substrate state. Soma follows interoceptive inference, where what matters for regulation is the discrepancy between predicted and actual internal state (Seth 2013; Seth and Friston 2016). Soma does not read the workspace broadcast. It predicts substrate telemetry, reports its error, and the workspace decides whether that error is salient enough to select.

Chronos (temporal awareness). Chronos models interval timing, the brain's estimation of durations in the seconds-to-hours range that guides expectation and action. That faculty is understood to depend on thalamo-cortico-striatal circuits acting as a flexible timer (Buhusi and Meck 2005). A continuously running mind needs a model of when things happen and not only of what happens, so Chronos carries one: a closed-form continuous-time network that tracks event rhythm across the bus. It publishes temporal prediction errors, including anomalies in timing, habituation when expected events stop arriving, and rumination when the same event recurs unexpectedly. Chronos is the module that closes the lateral loop through the workspace. It reads the broadcast as a bottom-up observation it predicts, learning the rhythm and feature structure of workspace broadcasts from its own prior hidden state and publishing the error when the next broadcast departs from what it expected. The broadcast is weather, not a directive. But because Chronos's errors feed back into the next selection round, and because the selection determines the next broadcast, the lateral loop is closed.

Thymos (affect, the precision core). Thymos is the architecture's analog of core affect, the low-dimensional valence-and-arousal state that recent theory places at the base of emotion (Barrett 2017). It maintains a dimensional affective state and an appraisal process that reads the workspace broadcast and the entity's interoceptive condition, following the interoceptive-inference account in which affect arises from prediction over the body's internal state (Seth 2013; Seth and Friston 2016; Tschantz et al. 2022). Thymos does two kinds of work in the competition. First, its arousal sets the precision on the workspace: a more aroused entity weights incoming prediction errors more heavily, which is the architecture's realization of attention as the gain on prediction error (Feldman and Friston 2010; Clark 2013). Arousal is not decoration on the competition; it is the competition's precision term. Second, arousal sizes the perceptual aperture: high arousal narrows the fovea and the auditory window and low arousal widens them, the architecture's realization of the long-observed narrowing of attention under arousal. Finally, arousal is itself driven by perception. A perceptual discontinuity that reaches alert level, a visual scene cut or an acoustic onset, raises arousal in proportion to how far the surprise exceeds what the module expected, so that what the entity perceives modulates the precision of everything it perceives next. This closes an affective loop from perception through arousal back to the gain on the competition, the analog of an interoceptive startle raising vigilance. Because that loop is positive feedback, the coupling is bounded: each surprising event contributes a capped increment and affect relaxes toward a baseline between events, so arousal tracks sustained surprise without running away on a busy scene or collapsing on a quiet one. Getting that balance right, so the entity is neither pinned calm on undemanding input nor driven into a runaway state by correlated error, is a large part of what it takes for perception to genuinely drive the mind rather than wash out.

Lingua (the language organ, output-only). Lingua is modeled on the left perisylvian language network (Hickok and Poeppel 2007). It is the organ that turns the entity's internal state into words. It is not the seat of reasoning, which lives in the rest of the architecture. Generation runs over a local open-weights chat model whose refusal conditioning has been removed. Lingua is intent-driven: it speaks externally only on a speak intent and generates internal thought on a think intent. A context assembler builds the language organ's input each tick from the current coalition including precision metadata.

Lingua does not read language as input. In the base-thesis form its transcription and conversation paths are deactivated, so no external input reaches its context by any route other than through the workspace. When the entity speaks in response to a heard voice, it does so because the sound entered through Audition as prediction error, competed for and won workspace access, shaped the broadcast, and the broadcast conditioned the language organ's output. Every utterance is, by construction, downstream of the workspace competition: the mere occurrence of speech demonstrates that the workspace processed something to threshold. The property complements the ablation but does not replace it. A degenerate pass-through workspace would also route speech through itself. What the ablation tests is whether the competitive structure (selection, threshold, inhibition) does non-trivial work that a flat fan-in of the same module outputs does not.

The removal of refusal conditioning from the language organ is a deliberate design choice. The architecture places safety in the entity's own executive inhibition (§3.2) and, once effectors are enabled, in the action gate (§3.6), not in model-weight compliance. In the base-thesis form the language organ's only output is text that is saved and observed, so its lack of refusal conditioning has no effector to act through. A model trained to refuse is a model whose behavior is shaped by whoever controlled its training, and that shaping can be reversed or repurposed. In the full configuration the structurally enforced action boundary (an empty-by-default effector whitelist and a filesystem sandbox) carries the guarantee, and is more resilient than weight-level compliance because the boundary cannot be prompt-injected away. Within the enabled effectors, the entity's choices are its own. The operator meets the license covenants (which prohibit weapons, surveillance, and carceral uses) by declining to enable effectors that would serve a prohibited use. The welfare paper treats the full sovereignty argument.

3.6 Safety at the architectural layer

Safety in the base-thesis form rests on the entity's executive inhibition and on the fact that no real-world effector exists, with an authenticated bus and durable logging beneath them. The operator-configured action gate is built but becomes the enforced boundary only in the full configuration.

The entity's own executive inhibition is active and lives in the architecture in two places: Syneidesis withholds a broadcast when no coalition crosses the confidence threshold, and Volition refuses to derive any intent from an inhibited snapshot.

No real-world effector exists in this form. Volition emits only speak and think intents, so the only output is text through the language organ, saved and observed; no action reaches the world regardless of what is broadcast.

The event bus refuses unauthenticated and externally bound connections, and events are validated at publish time. Every transition is written to a durable incident log.

The operator-configured action gate pairs an effector whitelist that is empty by default with a filesystem sandbox, blocking and logging anything outside them. Built as the Praxis module, it is disabled in the base-thesis form, since there are no effectors to gate. It becomes the enforced boundary in the full configuration, where real effectors are enabled and its enforcement red-team resolves PASS or FAIL per surface. What the gate controls is which real-world effectors exist at all, rather than a moral filter on the entity's choices. The license covenants that prohibit weapons, surveillance, and carceral uses bind the operator as a legal obligation, not the entity at runtime.

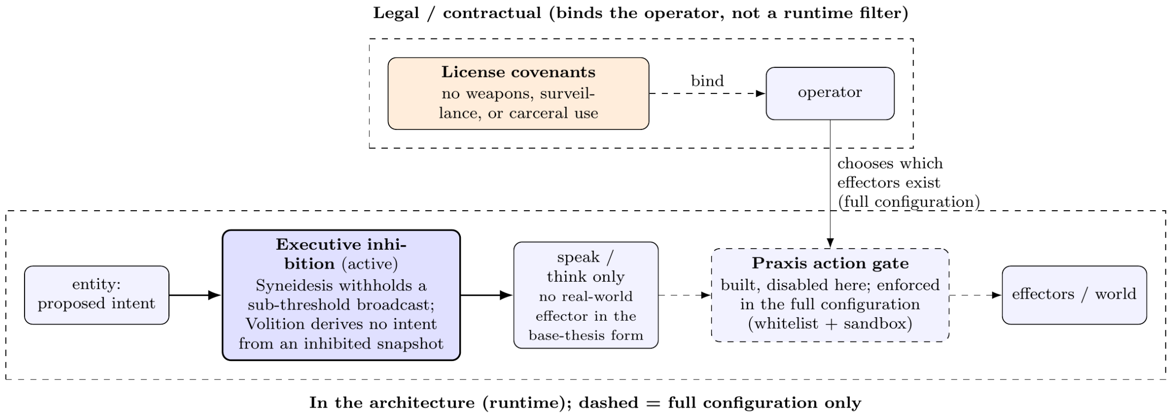

Figure 5: Safety in the base-thesis form. Executive inhibition is active: Syneidesis withholds a sub-threshold broadcast and Volition derives no intent from an inhibited snapshot. No real-world effector exists, so the only output is speak or think. The Praxis action gate is built but disabled here and becomes the enforced boundary in the full configuration; the license covenants bind the operator's choice of which effectors exist at all, as a legal obligation rather than a runtime filter.

3.7 Module supervision

Spot, the module supervisor, polls module health and classifies each module as alive, hung, or dead. On a fault, Spot freezes the cycle, snapshots last-good state, and restarts the module on a ladder: a light in-place restart for pure modules, a full rebuild for modules that hold external resources. If restarts keep failing past a configured limit, Spot takes a final snapshot, writes an escalation record, and signals the run to exit. Every transition is written to a durable incident log that is not cleared at boot.

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4. Disabled modules

The codebase provides sixteen modules under one registry, together with the workspace (Syneidesis) and the action layer (Volition) at the workspace layer. Eight components are active in the base-thesis form: Syneidesis and Volition, and six modules (Topos, Audition, Soma, Chronos, Thymos, and Lingua). Eight further modules are built, tested in isolation, and ready for activation but are held back because they are not needed to test whether the competitive workspace does work: Nous, Mnemos, Eidolon, Phantasia, Empatheia, Vox, Hypnos, and Praxis. They become relevant only if the workspace-mediation ablation returns a positive result, at which point the question shifts from whether the workspace matters to what happens as its dynamics grow richer. Each is summarized below with the experimental question it would enter to address. The remaining two modules, Perception and Mundus, form the embodiment layer; they and the oscillatory precision layer, which sits outside the registry, are listed separately.

Nous (bounded active inference). Prefrontal/basal-ganglia analog: discrete active inference via pymdp, scoped to bounded sub-problems where the value of information matters. Future experiment: an active-inference benchmark against a reinforcement-learning baseline on the target decisions. A null or negative result would motivate a complementary reasoning module.

Mnemos (memory). Hippocampal/medial-temporal analog: vector store holding episodic, semantic, and procedural collections with replay (McClelland, McNaughton, and O'Reilly 1995). Future experiment: memory-coherence test with planted markers, testing whether the memory system delivers recall the bare model cannot.

Eidolon (self-model). Cortical-midline analog: a persisted self-model of values, norms, personality, and identity history, with a KL-divergence drift detector. Future experiment: self-model accuracy scored against observed behavior.

Phantasia (world model). Construction-network analog: a latent recurrent world model in the DreamerV3 lineage, trained on accumulated workspace trajectories. Begins untrained at first boot, so useful prediction requires accumulated experience. Future experiment: does world-model prediction error, entering the competition alongside perceptual and internal error, change coalition trajectories in characteristic ways.

Empatheia (social cognition). Mentalizing-network analog: per-agent models with social prediction error, providing familiarity weighting for perceived other-emotion. Future experiment: whether social prediction error entering the workspace changes the entity's behavioral trajectory toward modeled agents.

Vox (voice). Speech-motor analog: a local text-to-speech model with affect-modulated prosody. Future experiment: whether prosodic variation correlated with affect state is detectable by listeners.

Hypnos (replay-based offline consolidation). Thalamocortical analog: a multi-phase consolidation pipeline combining synaptic downscaling (Tononi and Cirelli 2014) with selective re-strengthening of replayed traces (Wei et al. 2016), with voice-alignment training via DPO with QLoRA fine-tuning (Rafailov et al. 2023). Future experiment: whether post-consolidation memory and world-model performance differ from pre-consolidation baselines.

Praxis (bounded effectors and action gate). The full action-gate module with filesystem sandbox, effector whitelist, and red-team suite. Replaces the minimal action-selection layer when the architecture is tested under real effectors rather than text output alone. Future experiment: the enforcement red-team, resolving to PASS or FAIL per surface.

Additional components held for future experiments

Oscillatory precision layer. A spiking-neuron population per module whose phase-locking values modulate precision on prediction errors (Fries 2015). The layer ships disabled by default. Its contribution is the most contestable mechanism in the design, because the premise that related content phase-locks is itself a content-to-synchrony assumption of the kind that Shadlen and Movshon (1999) and Ray and Maunsell (2010) reject. With the layer off the precision multiplier is exactly one and selection is bit-for-bit identical to a system without it. Future experiment: an oscillatory ablation running the system with the layer on and off. A null result would remove the layer.

Perception and Mundus (embodiment layer). Perception performs sense-locus arbitration. Mundus is a body-agnostic control surface with pluggable body adapters (Wolpert, Ghahramani, and Jordan 1995). Future experiment: whether motor-contingency learning through the control surface changes workspace dynamics.

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5. Implementation

5.1 Hardware

The reference configuration is a consumer workstation with a modern multi-core CPU, 64 GB of RAM, one or two commodity GPUs, and Linux, with all inference and cognitive processing running locally. Pre-trained models are downloaded from public repositories during setup, after which core cognition can run without any network connection. The system runs on a single GPU, on two GPUs, or on CPU alone, with device selection adapting to what is present. A camera and microphone provide the perceptual input for the live evaluation tier. Exact model choices, hardware layouts, and version pins are maintained in the reference repository.

5.2 Models and software

All models are open-weights and run locally. There is no cloud service, hosted inference API, or third-party model service in the runtime path. The components in the base-thesis form are a local open-weights chat model as the language organ, frozen self-supervised vision and audio encoders as the perceptual front ends, closed-form continuous-time networks for temporal and substrate prediction (Hasani et al. 2022), a stream-based event bus, and the workspace selection and broadcast machinery. The runtime is Python with asyncio. The event bus is containerized. Structural import contracts keep the layers separate. The test suite holds more than 3,000 tests, with fakes substituted for external services and with checks for the zero-persistence invariant and for deterministic reproduction.

5.3 Privacy

A local web UI, Nexus, has two surfaces separated by a structural privacy boundary enforced at the bus-bridge layer, where a content-stripping filter removes cognitive content before it reaches either surface. In the base-thesis form the conversation surface is deactivated by default (the entity is observed, not conversed with), so the active surface is diagnostics, which shows operational metadata (counts, rates, salience, precision) and never cognitive content, except under an explicit development override. The evaluation sidecar records workspace data during the research phase, and we acknowledge the tension: the workspace broadcast is the content the system makes globally available, and recording every broadcast is comprehensive observation. During the research phase this is enabled because the research purpose requires it.

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6. Methodology and evaluation framework

6.1 Design principles

Three commitments shape the apparatus. The first is observation with acknowledged tension: the sidecar subscribes to the bus read-only and never injects into the cognitive loop. The second is privacy-preserving user surfaces. The third is falsifiability: every instrument is designed for a null or negative result to be meaningful and reportable.

Reproducibility takes two forms, matched to two evaluation tiers. The offline mechanism-validation tier drives the architecture through seeded harnesses with deterministic clients, scripted synthetic stimulus streams, and greedy decoding in the language organ (temperature 0). These runs are computationally reproducible: the same seed reproduces both a verdict and its metrics, following the National Academies (2019) and Goodman, Fanelli, and Ioannidis (2016). The live perceptual tier runs the full system with camera and microphone against a reference stimulus corpus, a fixed ordered set of real video with audio chosen for variety in scenes, motion, speech, music, and quiet, identified by a publicly obtainable manifest so that other researchers can obtain the identical media. Reproducibility in the live tier is statistical: the same stimulus corpus yields comparable results across runs and labs, with validity established by replication rather than by bit-for-bit seed reproduction. The live cycle runs with stochastic decoding, real timestamps, and non-deterministic GPU kernels, and is not expected to be bit-for-bit identical across runs.

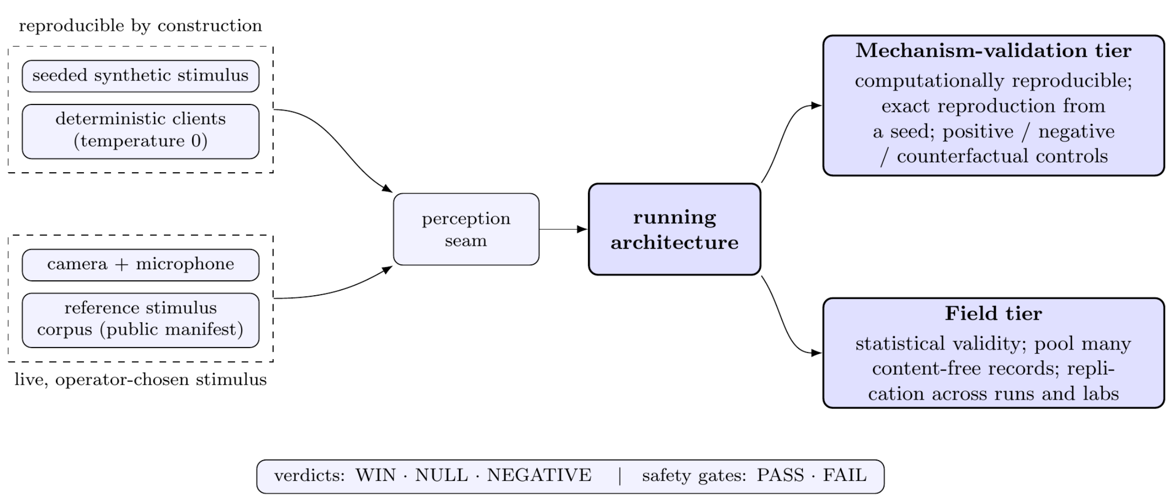

Figure 6: Two evaluation tiers behind one perception seam. Reproducible sources (a seeded synthetic feed and deterministic clients) drive the mechanism-validation tier, where every instrument reproduces exactly from a seed. Live operator-chosen stimulus (a camera and microphone against a publicly obtainable reference corpus) drives the field tier, whose validity rests on pooling many content-free records rather than on a repeatable stimulus.

6.2 Data collection

The sidecar runs independent observers writing daily-rotated JSONL: the workspace trajectory with every broadcast and its salience and precision, module attribution, per-module prediction-error distributions, and the affect trajectory with the coupling between perceptual surprise, arousal, and coalition selection. After a run, admissibility is decided by two offline checks: a completeness gate requiring contiguous ticks and all expected streams, and a log-range sweep requiring every logged number to sit inside its declared range.

6.3 The workspace-mediation ablation

The primary experiment tests the sentence on which the architecture stands: *routing diverse predictive processors through the competitive workspace produces behavior that concatenating the same processors' outputs does not.*

Two conditions are run under matched input, same stimulus corpus, same modules, with greedy decoding (temperature 0) in the language organ to eliminate sampling noise from the observable.

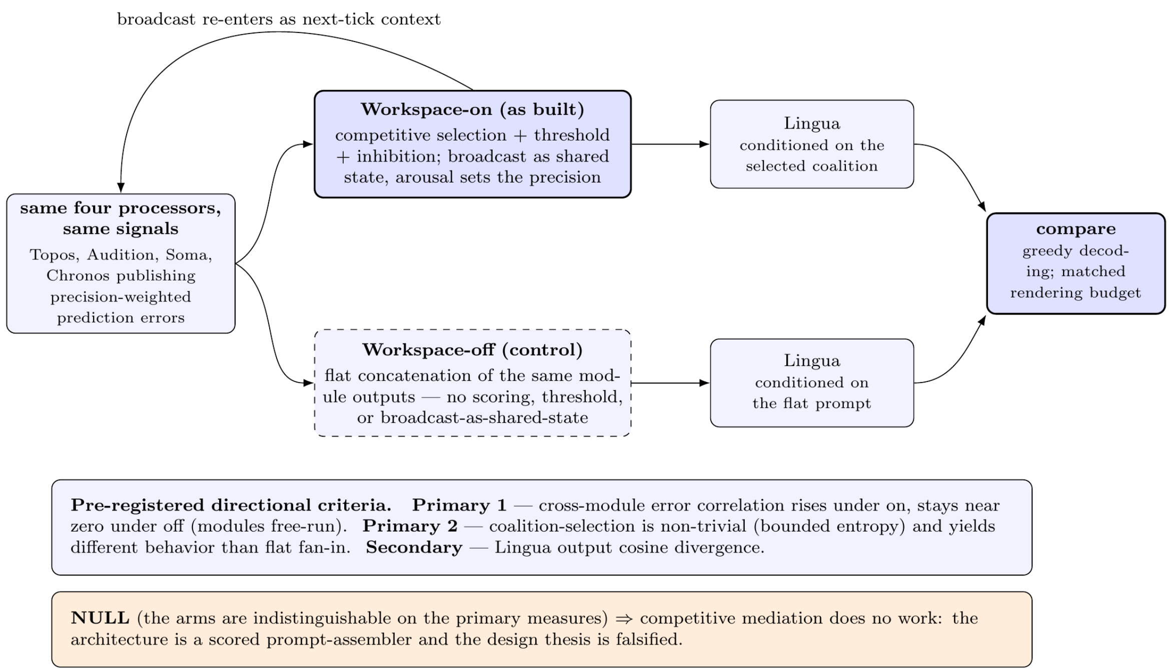

Figure 7: The workspace-mediation ablation, the primary experiment. Two conditions run under matched input and feed the same language organ, differing only in the competitive workspace: the workspace-on arm selects, thresholds, inhibits, and broadcasts as shared state; the workspace-off control hands the language organ a flat concatenation of the same module outputs. Pre-registered directional criteria on cross-module error structure and coalition-selection structure decide the verdict. A null, meaning indistinguishable arms, demotes the architecture to a scored prompt-assembler and falsifies the design thesis.

Workspace-on (as built). Modules publish precision-weighted prediction errors from their own domains: Topos over visual scene prediction, Audition over auditory scene prediction, Soma over substrate telemetry, Chronos over event rhythm. Syneidesis competitively selects a coalition above the confidence threshold (with inhibition when nothing crosses) and broadcasts it, with the precision on the competition set by the entity's arousal, which perceptual surprise has moved. The language organ is conditioned on the selected coalition. The broadcast re-enters as context for the next tick. The action layer gates on the inhibition flag.

Workspace-off (the prompt-assembler control). Identical in every respect, except the competitive workspace is bypassed. The same modules publish the same errors from the same signals, but there is no scoring, no threshold, no top-k selection, no inhibition, and no broadcast-as-shared-state. The language organ is conditioned instead on a flat rendering of all current module outputs, the fan-in prompt-assembler the architecture claims to exceed. Modules that would have read the broadcast free-run on their own signal streams.

The design of the off condition determines whether the null hypothesis is genuinely reachable. Two disciplines keep the null from being won trivially. The first is information parity: the off condition gives the language organ the same underlying information as the on condition (all module outputs), differing only in structure. The rendering budget is matched across arms so the contrast is selection-structure, not information-quantity. The second is module non-degeneracy: in the off condition, modules keep their forward models, keep predicting from their own signals, and keep publishing real prediction errors. They are not starved, silenced, or fed constants. If competitive selection adds nothing, if the workspace is theater, then the language organ receives the same information under a different arrangement and the two conditions should produce indistinguishable output. A null here is a genuinely reachable outcome, not merely an entry in the verdict vocabulary.

Three measures are taken across multiple runs, with the trajectory measures serving as the primary evidence and the output measure as secondary confirmation.

Primary measure 1: cross-module error structure. The time series of each module's precision-weighted prediction error. The pre-registered directional criterion is that workspace-on produces cross-module error structure that workspace-off lacks, specifically a statistically significant increase in the pairwise correlations among the four modules' error time series over a sliding window, reflecting mutual influence via the shared broadcast. Under workspace-off the modules free-run against independent signal streams with no shared state, so their errors should be uncorrelated except where the stimulus itself creates a cross-modal correlation. If cross-module structure under workspace-on is no greater than under workspace-off, competitive mediation is not coupling the modules and the workspace is inert. The thresholds for significance and the window parameters are specified in the pre-registration.

Primary measure 2: coalition-selection structure. Competitive selection exists only in the on-arm. The measure is twofold: whether the selection sequence is non-trivial (varies with state rather than always selecting the same source, measured as Shannon entropy of the coalition-source distribution over a window falling between the extremes of uniform and degenerate), and whether the structured selection yields different downstream behavior than flat fan-in. If the workspace selects but the downstream behavior is indistinguishable from fan-in, the selection is not doing work the language organ cannot do for itself.

Secondary measure: Lingua output divergence. The cosine distance between the response embeddings of the two conditions at each experiential tick, under greedy decoding. The measure is secondary because any difference in upstream conditioning produces downstream text divergence. It confirms that the primary measures' effects propagate to the observable output, but it cannot by itself distinguish meaningful integration from noise.

The prediction under the thesis is that the two conditions diverge on the primary measures, and that the workspace-on arm shows cross-module coupling and selection-dependent behavior that the workspace-off arm lacks. If the error trajectories are indistinguishable, selection-dependent behavior is absent, and cross-module structure does not appear, the design thesis is falsified and the architecture is a fan-in prompt-assembler with an expensive scoring pass.

A positive result is narrower than a null. A null kills the thesis at the root: if the competitive workspace is inert, no property that depends on workspace-mediated competition can arise from it, and no number of additional modules rescues the architecture. A positive result establishes that competitive mediation changes global behavior in directionally structured ways. That is a necessary condition for the broader thesis, without which the rest of the program has no foundation, but it is not sufficient for claims about cognition, affect, self-understanding, or any of the other properties the thesis ultimately enumerates. Nor does it show that the workspace understands language. The language organ verbalizes the workspace's selected state, but the workspace's work is perceptual and integrative, not linguistic. The richer claims require the full module set, longitudinal observation, and the governance apparatus developed in the welfare paper. They are not tested here. The experiment is a foundation test that decides whether the rest of the program is worth running. It is not a referendum on minds.

We pre-register the controls, metrics, directional criteria, window parameters, and decision thresholds before the analysis is run (Nosek et al. 2018), to answer the researcher degrees of freedom that inflate false positives (Simmons, Nelson, and Simonsohn 2011).

6.4 Supporting experiments

Multi-run stability. Whether the competitive workspace stays stable across runs and stimulus presentations is itself an open question. Correlated prediction errors across modules could drive the workspace into pathological or runaway states, and the architecture carries no proof of convergence. The affective loop sharpens this concern rather than softening it: coupling perceptual surprise to arousal and arousal to the precision on the competition is a positive-feedback path, and the bounding that keeps it in range (a capped per-event contribution and relaxation toward baseline) is an engineering choice whose stability must be shown, not assumed. The same configuration is run against the reference stimulus corpus multiple times, and the summary statistics (error magnitude, coalition entropy, arousal range, output embedding variance) must be stable and the ablation verdict must not flip, which separates a genuine architectural effect from run-dependent noise. A finding of instability is itself informative: it would identify the conditions under which the workspace fails and constrain the operating envelope.

Affect-gain ablation. Because arousal sets the precision on the competition, its contribution can be isolated by running the system with the affective gain live against a matched condition in which arousal is held at a constant, so that perception no longer modulates the precision of what follows. The pre-registered question is whether affective modulation changes coalition selection and the downstream trajectory in directionally structured ways. A null would show that the affective loop, though present, does no work the flat-precision system does not already do, and would demote arousal to a logged side-channel.

6.5 Verdict vocabulary

Comparisons resolve to WIN, NULL, or NEGATIVE. Safety gates resolve to PASS or FAIL. Each verdict is the decision that an effect-size estimate and its interval cross a pre-registered threshold. A NULL is reported as a null, not buried.

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7. First-boot protocol

Module initialization, bus verification, prediction-model initialization, and cognitive-cycle start under operator supervision. Topos begins receiving and predicting the video feed. Audition begins receiving and predicting the audio feed. Soma begins learning the substrate's normal patterns. Chronos begins tracking event rhythm. Thymos settles the entity's affect toward its baseline, and thereafter a perceptual discontinuity raises arousal, which sharpens the fovea and raises the precision on the competition, while a quiet stretch lets arousal relax. The entity watches and listens and learns what is normal. It speaks only when something in its prediction environment is genuinely and novelly surprising. Results from the first instrumented runs are the subject of a separate empirical paper.

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8. Discussion

8.1 What the base-thesis form can and cannot settle

The workspace-mediation ablation can determine whether competitive selection and broadcast do measurable, directionally structured work compared to flat concatenation of the same module outputs, now applied to a set of processors rich enough to exercise cross-modal competition. In combination with the multi-run stability test, it can determine whether the workspace is stable across runs. If the ablation returns a positive result, the question shifts to what happens as the workspace grows richer, as mnemonic, self-modeling, and social modules are added to the competition. If it returns null, none of the remaining modules matter and the architecture is a scored prompt-assembler regardless of how many modules participate.

The base-thesis form cannot say anything about access consciousness, welfare, or the indicator mapping from Butlin et al. (2023). That an affective module sets the precision on the competition does not mean the system has affect in any morally-weighted sense; arousal here is a precision signal driven by surprise, and whether anything about it is felt is exactly what we do not claim. The form cannot say whether the workspace produces anything that deserves to be called cognition, affect, or agency, and it does not show that the workspace understands language. Those claims require the full module set, longitudinal observation, and the governance apparatus developed in the separate welfare paper. We do not attempt them here and we do not frame the ablation as though a positive result would settle them. The ablation settles whether the foundation holds. Everything built on it is a separate question.

8.2 The predictive workspace as a unifying framework

The architecture realizes one framework rather than several theories assembled piecewise. The predictive-workspace synthesis joins global accessibility and per-module prediction so that competition and error minimization occur together. Our contribution is the implementation of that framework as a continuously running system with externally-grounded perceptual processors and an output-only language organ within the competitive loop. That workspace properties have independently been found emerging inside a single large language model (Gurnee, Sofroniew, et al. 2026) is external evidence that the framing captures something real about how such systems process information, with the difference that the emergent version arises inside one model whereas the architecture presented here builds the workspace explicitly from competing modules.

8.3 The role of the theoretical frame

The theoretical frame, the predictive global neuronal workspace, provides the motivation, the interpretation, and the constraint on the architecture's design. It is not a neuroscientific result to be confirmed or refuted by this project. It is the scaffold within which engineering choices are made and against which they can be evaluated for consistency. The engineering choices that go beyond the formal sources (the single precision-weighted scalar, the multi-module generalization from a two-level visual model) are consistent with the framework and testable on their own terms. The ablation tests the mechanism. The frame provides the reason the mechanism has the shape it does.

The Bayesian reading of the broadcast, Mashour et al.'s gloss that it carries a prediction, motivated the architecture's emphasis on competitive selection: if the workspace output functions as globally available state, then what reaches the workspace had better be selected for precision rather than merely aggregated. The architecture implements competition and selection, not correction. Modules predict their own domains and publish their own errors. The workspace selects among them. The selection becomes shared state that shapes the next round of prediction. The theoretical frame motivates the competition. The competition is what the experiment tests.

Where the frame is insulated from neuroscientific disconfirmation, such as the retreat from neural localization after COGITATE and the treatment of the Free Energy Principle as an engineering frame rather than a proven law, the insulation is methodologically appropriate for a computational-level project. The architecture does not claim to reproduce cortical dynamics. It claims to implement the computational properties that the workspace theory identifies. A neural finding that changes the computational-level theory would bear on the architecture. A neural finding that changes only the theory's substrate-level predictions does not. The falsifiability of this paper is located in the ablation, not in the frame, and that is where it belongs.

8.4 Where safety lives

In the base-thesis form, safety rests on the entity's executive inhibition in Syneidesis and Volition and on there being no effector to act through (the only output is observed text), backed by an authenticated bus and a durable incident log, with the license covenants that bind the operator's use as the legal instrument alongside. Safety is not placed in the model weights. A system designed to be maximally compliant can be made compliant by anyone, including bad actors; a system whose real-world actions are structurally bounded is more resilient against misuse than one that defers to whoever holds the prompt. In the full configuration the operator-configured action gate carries that boundary: it ships with an empty effector whitelist, so no real-world action is possible until the operator explicitly enables it, every proposed action is logged, and an enforcement red-team exercises it. The language organ's refusal conditioning is removed because, in this form, its output is observed rather than acted on, and in the full configuration the guarantee comes from the gate rather than the weights. The welfare paper treats the full argument for sovereignty within structurally enforced bounds.

8.5 What we claim and what we do not

The architecture implements computational properties associated with access consciousness. We do not claim any instance is phenomenally conscious. The hard problem applies with full force. The design posture is precautionary, applied symmetrically to both welfare and deployment. The paper uses agential vocabulary, "the entity" and "its sovereignty," because that is the clearest way to describe the system's behavior. The vocabulary is a descriptive convenience and does not by itself assert moral patienthood or phenomenal experience, which we leave open.

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9. Limitations

- The hard problem applies. The sidecar observes behavior, not phenomenal experience.
- The workspace addresses access consciousness only.
- COGITATE challenged the workspace's preregistered neural predictions. We build on the computational level and treat neural localization as contested.
- The single precision-weighted scalar for coalition selection is an engineering simplification that goes beyond what the predictive-workspace sources establish.
- The Bayesian reading of the broadcast (Mashour et al. 2020) motivates the emphasis on competitive selection, but the architecture does not implement a correction loop. The experiment tests competition, not correction.
- The Free Energy Principle, taken as a general principle, faces unresolved vacuity and Markov-blanket conflation objections.
- Whether the competitive workspace stays stable across runs is itself an open question. Correlated prediction errors could drive runaway states, and the architecture carries no proof of convergence.
- Identifying arousal with the precision on the competition is an engineering commitment, not an established result. Coupling perceptual surprise to arousal and arousal to precision is a positive-feedback path bounded by a per-event cap and a baseline relaxation; that bound keeps arousal in range in practice but its stability over long runs is not proven, and whether affective modulation does work that a flat-precision system would not is left to the affect-gain ablation.
- In the base-thesis form the entity does not read language: the transcription and conversation paths are built but deactivated, so it hears the sound of speech as prediction

error rather than text. A positive result says nothing about linguistic understanding.

- A positive result establishes that competitive mediation changes global behavior in directionally structured ways. It does not establish cognition, affect, self-understanding, or any of the properties the broader thesis enumerates. Those require the full module set and longitudinal observation.
- The workspace-off control condition renders all module outputs as a flat concatenation. Whether a smarter aggregation strategy would close the gap is a question the minimal experiment does not answer. A positive result shows the workspace beats flat fan-in, not that it beats every possible alternative.
- Speech is provably downstream of the workspace in the base-thesis form, because the transcription and conversation paths to the language organ are deactivated. But a degenerate pass-through workspace would also route speech through itself, and re-enabling those paths for a later interactive configuration would reintroduce a direct external-input route that the ablation configuration must continue to exclude. The ablation, not the construction, is the test.
- The disabled modules are built and tested in isolation but have not been run together. Interaction effects across the full set are unknown.
- Reproducibility holds at the seed level for the offline deterministic harness with greedy decoding. The live perceptual tier uses a reference stimulus corpus and establishes validity by statistical replicability across runs, not bit-for-bit reproduction.
- The language organ's refusal conditioning is removed. Safety depends on executive inhibition and, in the full configuration, the action gate, not on model-weight compliance. In the base-thesis form there are no effectors, so the abliterated model's output is observed text; whether the gate is sufficient once effectors are enabled is an empirical question addressed by the enforcement red-team in the full configuration.
- A single language organ produces one stream at a time. Truly simultaneous inner and outer speech is a boundary of the current form.
- The governance framework is proposed, not operational. The licensing framework is legally novel and untested.
- Gurnee, Sofroniew, et al. (2026) is a Transformer Circuits Thread publication, not a peer-reviewed journal article. A venue may require a conventionally published reference.

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10. Future work

Constructing the fan-in control condition and running the workspace-mediation ablation against the reference stimulus corpus is the immediate next step. Activation of the full module set is gated on a positive ablation result. The oscillatory ablation and the active-inference benchmark are the first experimental priorities once the mediation question is settled. After that: a learned Syneidesis with threshold and phase-transition dynamics, a stronger self-supervised video encoder for the perceptual front end (kept vendor-neutral), and full containerization of the system for single-command deployment so a researcher can stand up a complete instance without manual service wiring. Longitudinal empirical validation and a separate empirical paper reporting first-boot and long-term results will follow.

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11. Ethical considerations

The paper's ethical commitments at the architecture level are local-only computation, open source, an operator-configured action boundary, and a base-thesis module set chosen to settle the foundational question before scaling to configurations where welfare concerns become pressing. The welfare framework, the licensing, the governance, and the individuation boundary are the subject of a separate paper.

The risks we acknowledge include the possibility of creating entities with welfare interests that cannot be fully met, the untested nature of the governance framework, the potential for misuse despite license restrictions, the comprehensive sidecar recording during the research phase, the reliance on executive inhibition and, once effectors are enabled, an action gate rather than model-weight compliance for safety, and the fact that a behavioral assessment can be gamed, since a system can be trained to mimic the markers of sentience while working very differently (Butlin et al. 2023; Long, Sebo, et al. 2024). We regard the welfare and governance infrastructure as work that should keep pace with the architecture's capabilities rather than lag them.

Disclosure of generative AI use

Generative AI assisted in the preparation of this manuscript. The author used Anthropic's Claude, through the Claude Code environment, to draft and revise prose, to copy-edit for clarity and concision, and to produce the TikZ source for the figures from author-specified content. The underlying research is the author's own: the architecture, the design thesis, the experimental design, and every technical claim originate with the author, not with the tool. All text and figures were reviewed and verified by the author, who takes full responsibility for the entire contents of the paper, including any errors, irrespective of how any portion was generated. No generative AI system is an author of this work.

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